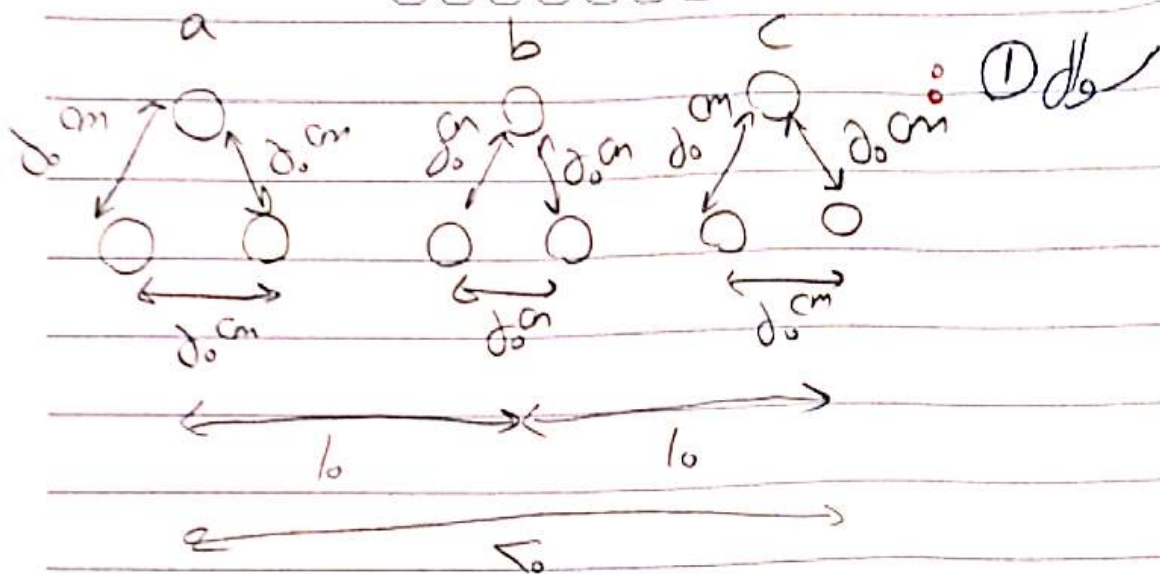


" آزمون برزری ۱۴۰۲/۱۰/۲۰ " " کویز سوم تحلیل "

Date: _____

Sat Sun Mon Tue Wed Thu Fri

Subject: _____



$$r' = 0.042 \text{ feet} = 0.042 \text{ feet} \times \frac{1 \text{ m}}{1 \text{ feet}} \times \frac{12 \text{ in}}{1 \text{ m}} \times 1.25 = 0.063 \text{ m}$$

$$= 0.018 \text{ m} \Rightarrow GMR = \sqrt[4]{r' d' d' d'} = \sqrt[4]{r' d' d' d'} = \sqrt[4]{0.018 \times 0.018 \times 0.018 \times 0.018}$$

$$= \sqrt[4]{0.0012} = 0.122 \text{ m} = GMR$$

$$D_{eq} = \sqrt{D_{ab} D_{ac} D_{bc}} = \sqrt{10 \times 10 \times 10} = 10 \sqrt{3}$$

$$\Rightarrow L = 2 \times 10^{-4} \ln \frac{D_{eq}}{GMR} = 2 \times 10^{-4} \ln \frac{10 \sqrt{3}}{0.122} = 2 \times 10^{-4} \ln(17.45)$$

$$= 2 \times 10^{-4} \times 2.85 \approx 1.12 \times 10^{-3} \frac{\text{H}}{\text{m}} \times \frac{1 \text{ km}}{1000 \text{ m}} = 1.12 \times 10^{-6} \frac{\text{H}}{\text{km}}$$

$$\Rightarrow X_L = \omega L_{\text{km}} = 2\pi f L_{\text{km}} = 2\pi \times 60 \times 1.12 \times 10^{-6} \approx 229.99 \times 1.12 \times 10^{-6} \approx 0.257 \frac{\Omega}{\text{km}}$$